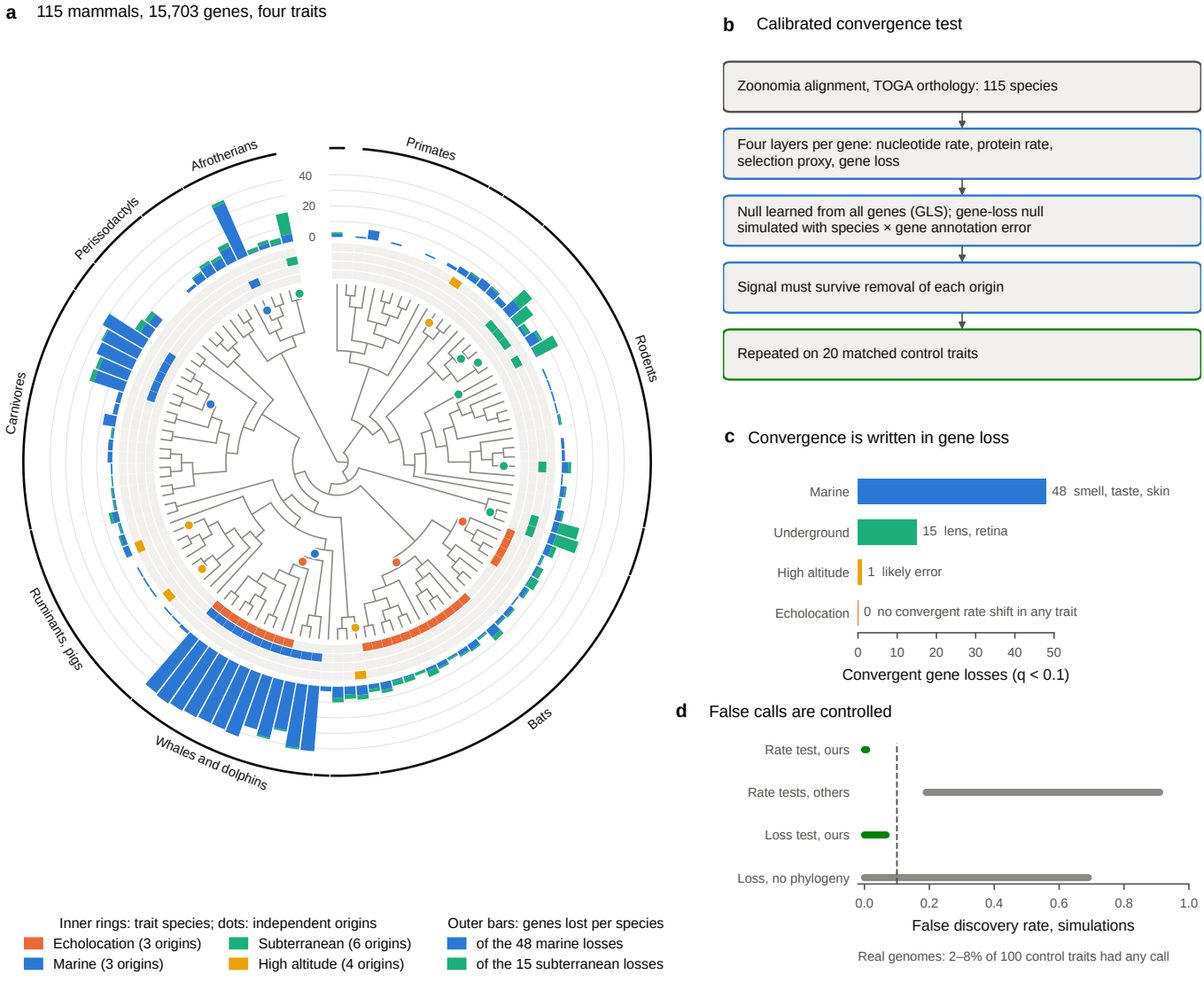

a
115 mammals, 15,703 genes, four traits
Primates
Rodents
Bats
Whales and dolphins
Ruminants, pigs
Carnivores
Perissodactyls
Afrotherians
40
20
0
Inner rings: trait species; dots: independent origins
Echolocation (3 origins)
Marine (3 origins)
Subterranean (6 origins)
High altitude (4 origins)
Outer bars: genes lost per species
of the 48 marine losses
of the 15 subterranean losses
b
Calibrated convergence test
Zoonomia alignment, TOGA orthology: 115 species
Four layers per gene: nucleotide rate, protein rate, selection proxy, gene loss
Null learned from all genes (GLS); gene-loss null simulated with species × gene annotation error
Signal must survive removal of each origin
Repeated on 20 matched control traits
c
Convergence is written in gene loss
Marine
48 smell, taste, skin
Underground
15 lens, retina
High altitude
1 likely error
Echolocation
0 no convergent rate shift in any trait
0
10
20
30
40
50
Convergent gene losses (q < 0.1)
d
False calls are controlled
Rate test, ours
Rate tests, others
Loss test, ours
Loss, no phylogeny
0.0
0.2
0.4
0.6
0.8
1.0
False discovery rate, simulations
Real genomes: 2–8% of 100 control traits had any call